

CS_5080_Lecture_10

Thursday, February 26, 2026 4:45 PM

Tabular TD(0) for estimating v_π

Input: the policy π to be evaluated
Algorithm parameter: step size $\alpha \in (0, 1]$
Initialize $V(s)$, for all $s \in \mathcal{S}^+$, arbitrarily except that $V(\text{terminal}) = 0$
Loop for each episode:
 Initialize S
 Loop for each step of episode:
 $A \leftarrow$ action given by π for S
 Take action A , observe R, S'
 $V(S) \leftarrow V(S) + \alpha[R + \gamma V(S') - V(S)]$
 $S \leftarrow S'$
 until S is terminal

Sarsa (on-policy TD control) for estimating $Q \approx q_\pi$

Algorithm parameters: step size $\alpha \in (0, 1]$, small $\varepsilon > 0$
Initialize $Q(s, a)$, for all $s \in \mathcal{S}^+, a \in \mathcal{A}(s)$, arbitrarily except that $Q(\text{terminal}, \cdot) = 0$
Loop for each episode:
 Initialize S
 Choose A from S using policy derived from Q (e.g., ε -greedy)
 Loop for each step of episode:
 Take action A , observe R, S'
 Choose A' from S' using policy derived from Q (e.g., ε -greedy)
 $Q(S, A) \leftarrow Q(S, A) + \alpha[R + \gamma Q(S', A') - Q(S, A)]$
 $S \leftarrow S'; A \leftarrow A'$
 until S is terminal

If we want
just evaluate
or obtain
 $q(s, a)$ values

Sarsa (on-policy TD control) for estimating $Q \approx q_*$

Algorithm parameters: step size $\alpha \in (0, 1]$, small $\varepsilon > 0$
Initialize $Q(s, a)$, for all $s \in \mathcal{S}^+, a \in \mathcal{A}(s)$, arbitrarily except that $Q(\text{terminal}, \cdot) = 0$
Loop for each episode:
 Initialize S
 Choose A from S using policy derived from Q (e.g., ε -greedy)
 Loop for each step of episode:
 Take action A , observe R, S'
 Choose A' from S' using policy derived from Q (e.g., ε -greedy)
 $Q(S, A) \leftarrow Q(S, A) + \alpha[R + \gamma Q(S', A') - Q(S, A)]$
 $S \leftarrow S'; A \leftarrow A'$
 until S is terminal

Q-learning (off-policy TD control) for estimating $\pi \approx \pi_*$

Algorithm parameters: step size $\alpha \in (0, 1]$, small $\varepsilon > 0$
 Initialize $Q(s, a)$, for all $s \in \mathcal{S}^+$, $a \in \mathcal{A}(s)$, arbitrarily except that $Q(\text{terminal}, \cdot) = 0$
 Loop for each episode:
 Initialize S
 Loop for each step of episode:
 Choose A from S using policy derived from Q (e.g., ε -greedy)
 Take action A , observe R, S'
 $Q(S, A) \leftarrow Q(S, A) + \alpha [R + \gamma \max_a Q(S', a) - Q(S, A)]$
 $S \leftarrow S'$
 until S is terminal

Double Q-learning, for estimating $Q_1 \approx Q_2 \approx q_*$

Algorithm parameters: step size $\alpha \in (0, 1]$, small $\varepsilon > 0$
 Initialize $Q_1(s, a)$ and $Q_2(s, a)$, for all $s \in \mathcal{S}^+$, $a \in \mathcal{A}(s)$, such that $Q(\text{terminal}, \cdot) = 0$
 Loop for each episode:
 Initialize S
 Loop for each step of episode:
 Choose A from S using the policy ε -greedy in $Q_1 + Q_2$
 Take action A , observe R, S'
 With 0.5 probability:
 $Q_1(S, A) \leftarrow Q_1(S, A) + \alpha (R + \gamma Q_2(S', \arg\max_a Q_1(S', a)) - Q_1(S, A))$
 else:
 $Q_2(S, A) \leftarrow Q_2(S, A) + \alpha (R + \gamma Q_1(S', \arg\max_a Q_2(S', a)) - Q_2(S, A))$
 $S \leftarrow S'$
 until S is terminal

need RL approaches that work with long eps efficiently or tasks with no episodes

MC $\rightarrow$ state action $q_t | s$ or $q_t | s, a$

q_t vals are computed and used to update $V(s)$ or $q(s, a)$ values.

if an ep. is not completed we do not have real q_t values

vector of $V(s)$ values

s_1	0
s_2	0
$\vdots$	$\vdots$
s_n	0

initialize $V(s)$ values to 0

s' - est of rtn we may get

terminal state has value 0

TD(0) algorithm

$V_{\text{new}}(s) \leftarrow$ part comes from $V_{\text{old}}(s)$ +
 part coming from current actions
 get an est. of q_t from current actions

alpha
 α = percent of

$$\begin{aligned} &\leftarrow (1 - \alpha) V_{\text{old}}(s) + (\alpha \cdot \text{est of } q_t) \\ &\leftarrow (1 - \alpha) V_{\text{old}}(s) + \alpha [r + q_{t+1} | s'] \\ &\leftarrow (1 - \alpha) V_{\text{old}}(s) + \alpha [r + \gamma V(s')] \end{aligned}$$

γ = gamma

$$V(s) \leftarrow (1-\alpha)V(s) + \alpha[r + \gamma V(s')]$$

$$\text{Let } \alpha = 0.1$$

90% V_{old}

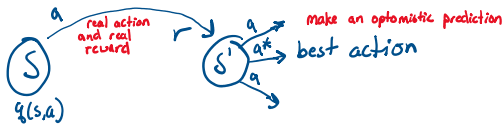
$$= 0.9$$

10% V_{old}

$$V(s) \leftarrow V(s) + \alpha [r + \gamma V(s') - V(s)]$$

| $\leftarrow$ TD error $\rightarrow$ |

Q-learning (Watkins, 1989)



$$Q(s,a) \leftarrow Q(s,a) + \alpha [r(s,a) + \gamma * \underset{\text{the best action that can be performed in state } s'}{\overset{\text{value for}}{Q(s',a)}} - Q(s,a)]$$

| $\leftarrow$ TD error $\rightarrow$ |

can quickly start to overestimate values

Q(s,a) table

States \ actions	a ₁	a ₂	...	a _n
s ₁				
s ₂				
...				
s _n				

Double Q-learning

two tables $Q_1(s,a)$ and $Q_2(s,a)$